

Digital Forensics Laboratory

EXPERIMENT NO: 01

DATE: August 9, 2026

TITLE: Comprehensive Digital Evidence Acquisition and Command-Line Analysis using FTK Imager, WinPmem, and Volatility

1. AIM

To perform a complete forensic acquisition and analysis lifecycle by imaging non-volatile storage using AccessData FTK Imager, capturing live volatile memory (RAM) using the WinPmem command-line utility, and analyzing the extracted memory dump using Volatility 3 commands via the PassMark Volatility Workbench.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tools:**
 - AccessData FTK Imager (GUI for non-volatile physical drive imaging)
 - WinPmem (winpmem_mini_x64_rc2.exe) (CLI tool for volatile memory capture)
 - Volatility 3 Framework (vol.exe / PassMark Volatility Workbench V3.0)
- **Hardware:** Evidence PC/Workstation, Target Storage Drive (hp v221w 16GB USB Device).

3. THEORY

A thorough digital forensic investigation requires collecting both data at rest and data in motion using strict operational procedures.

- **Non-Volatile Memory Acquisition:** AccessData FTK Imager creates mathematically verifiable, bit-by-bit physical images of storage devices (saved as Raw/dd) without altering the original evidence.

- **Volatile Memory (RAM) Acquisition:** Command-line utilities like **WinPmem** load a specialized kernel driver (pme98A2.tmp) to read physical memory directly. Executing this via CMD writes active processes, network connections, and decrypted keys to a raw dump file (.raw).
- **Volatile Memory Analysis (CLI): Volatility 3** is a framework for analyzing RAM dumps. By passing specific parameters and plugins (e.g., -f <image_path> windows.pslist.PsList) into the command line or workbench, investigators can reconstruct the system's live state and identify suspicious processes.

4. PROCEDURE

Part A: Acquiring Non-Volatile Memory (Disk Image) using FTK Imager

Step 1: Select the Source Type

Open FTK Imager as an Administrator. Navigate to **File > Create Disk Image**. In the Select Source dialog, choose **Physical Drive**.

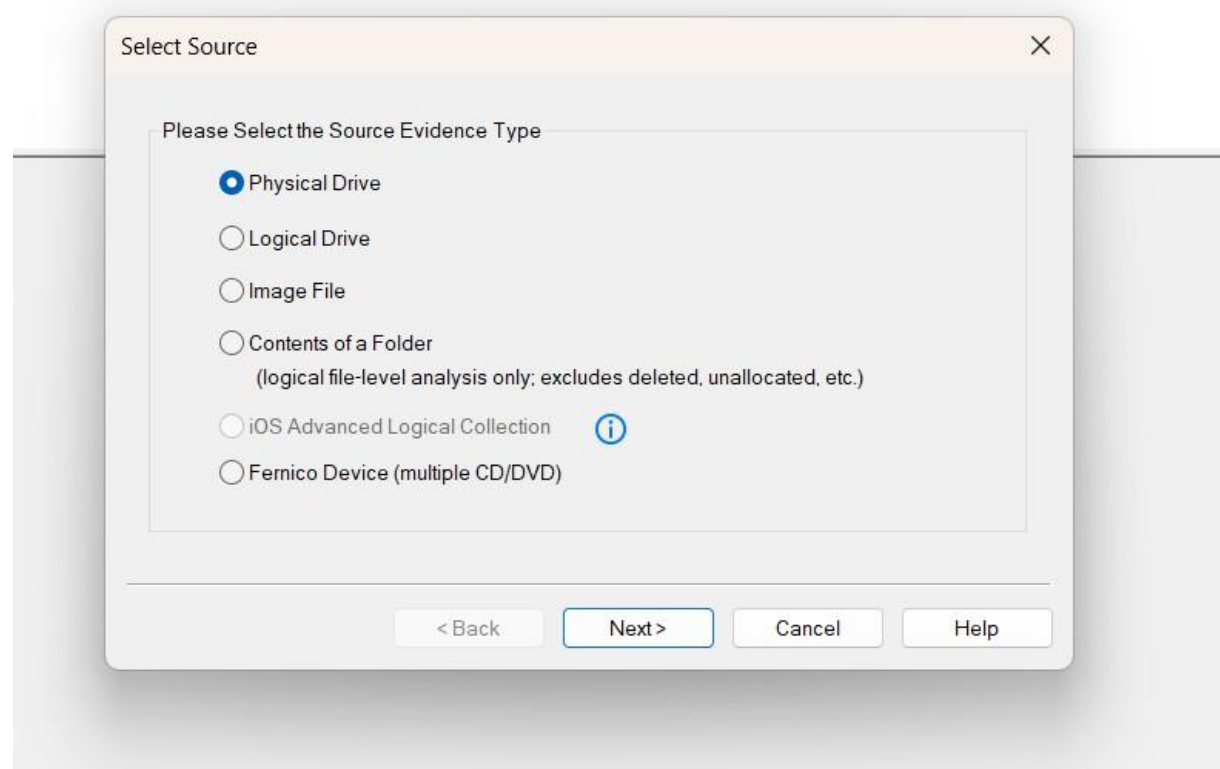


Figure 1: Selecting Physical Drive as the source evidence type

Step 2: Select the Specific Drive

Select the target evidence drive (\\.\PHYSICALDRIVE1 - hp v221w USB Device [16GB USB]) from the dropdown.

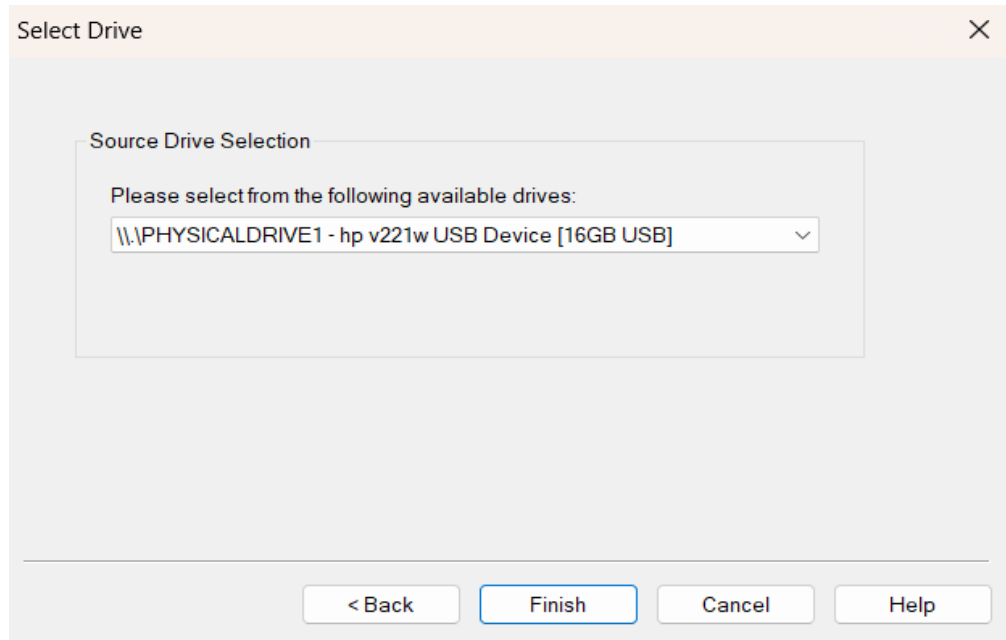


Figure 2: Selecting the target USB device for acquisition

Step 3: Choose the Image Type

Select **Raw (dd)** as the destination image type to create a pure bitstream copy.

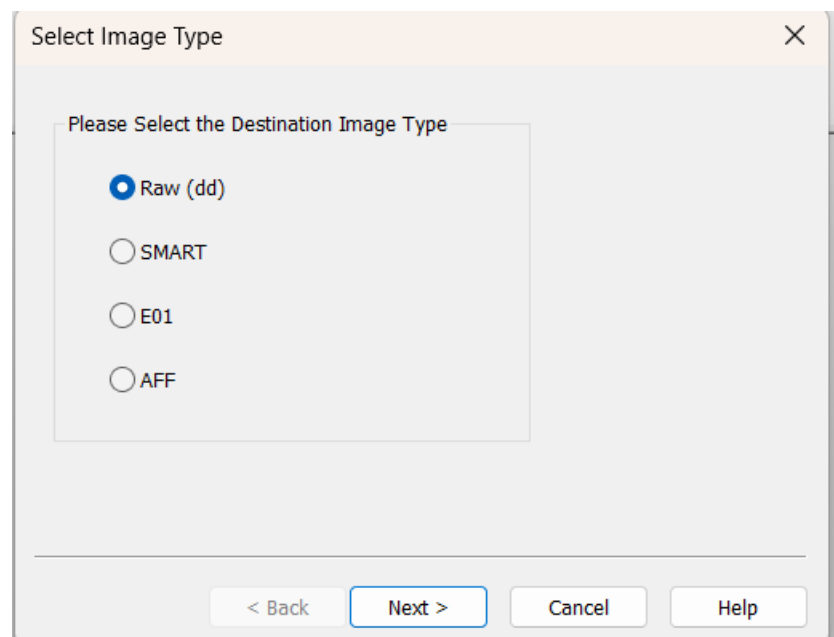


Figure 3: Configuring the destination image format to Raw (dd)

Step 4: Enter Evidence Item Information

Provide the standardized case details to maintain the chain of custody:

- **Case Number:** 99240040210
- **Evidence Number:** 1
- **Unique Description:** Examine
- **Examiner:** sadiq

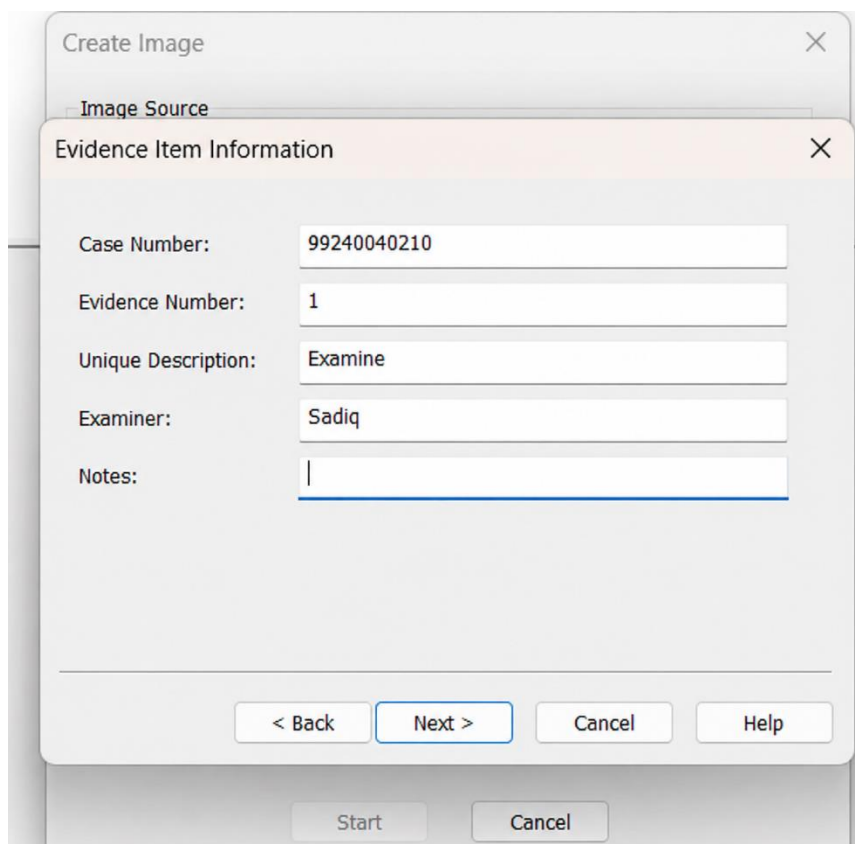


Figure 4: Entering the forensic case and examiner details

Step 5: Configure Image Destination

Set the destination folder to A:\DF\Digital_Forensics_Lab\Experiment-01-FTK-Imager\Results and the filename to Result. Set the Fragment Size to 20 MB and check **Verify images after they are created**.

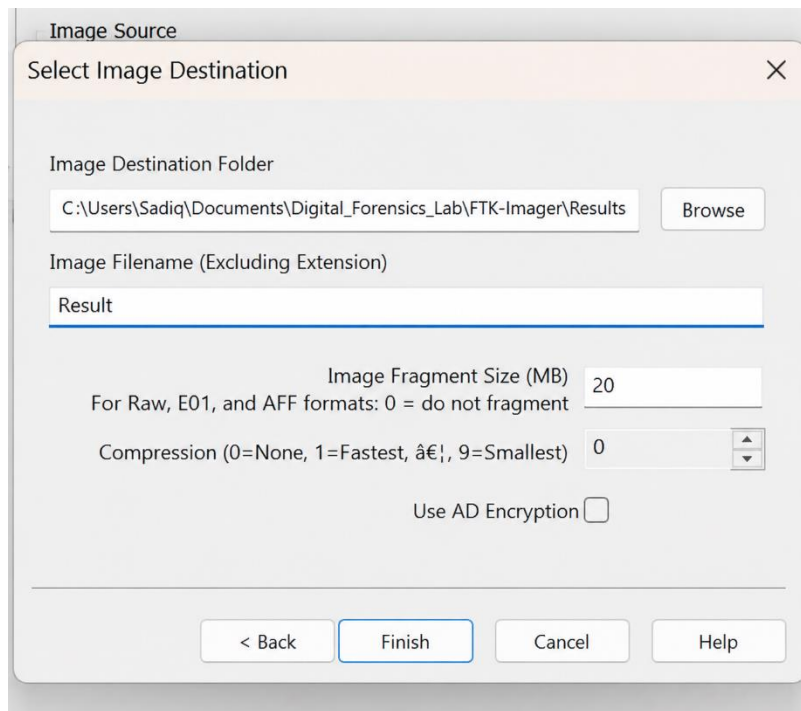


Figure 5: Setting the destination path, filename, and fragmentation size

Step 6: Monitor Acquisition Progress & Hash Verification

Click **Start**. Once complete, FTK Imager hashes the original drive and the newly created image. Review the Drive/Image Verify Results.

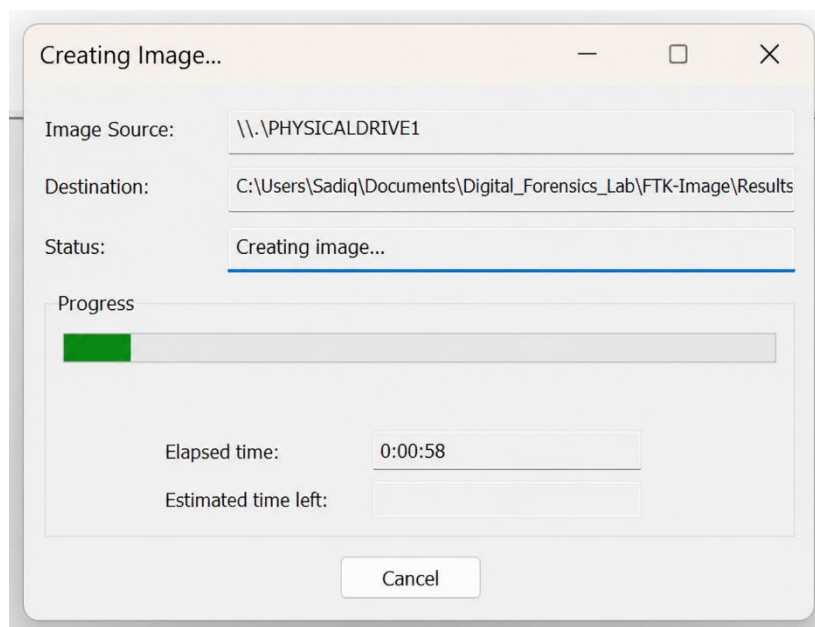


Figure 6: Disk imaging progress tracking

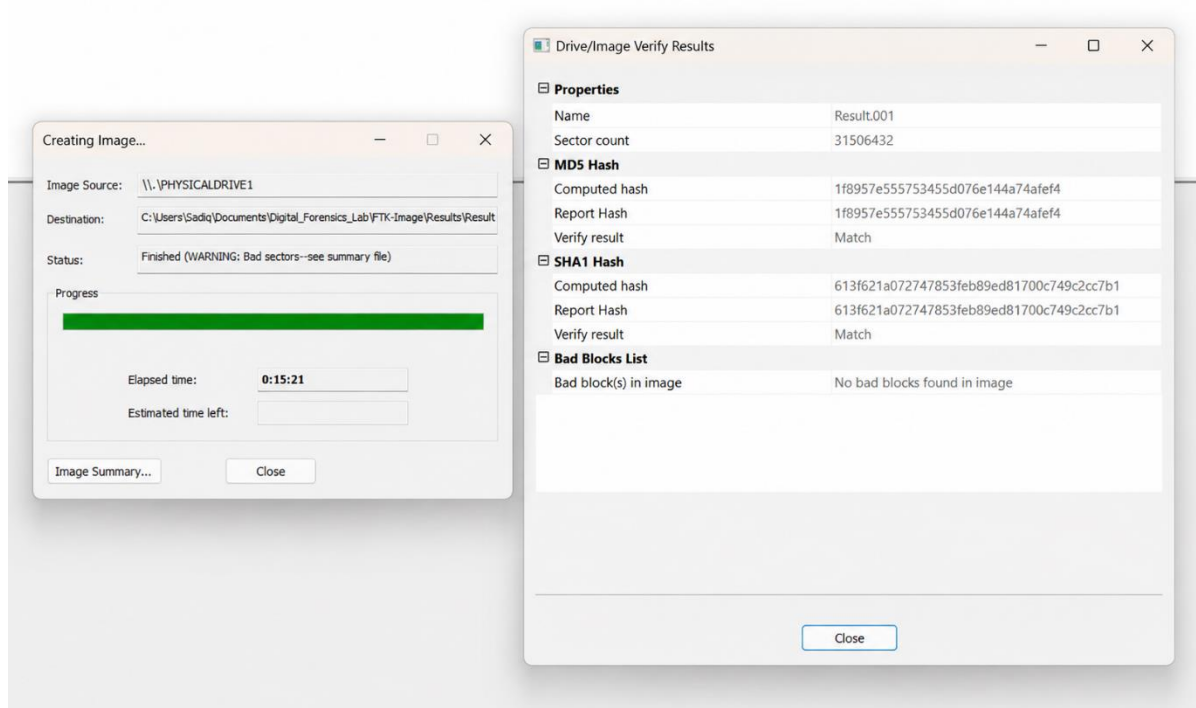


Figure 7: Successful verification of MD5 and SHA1 hashes

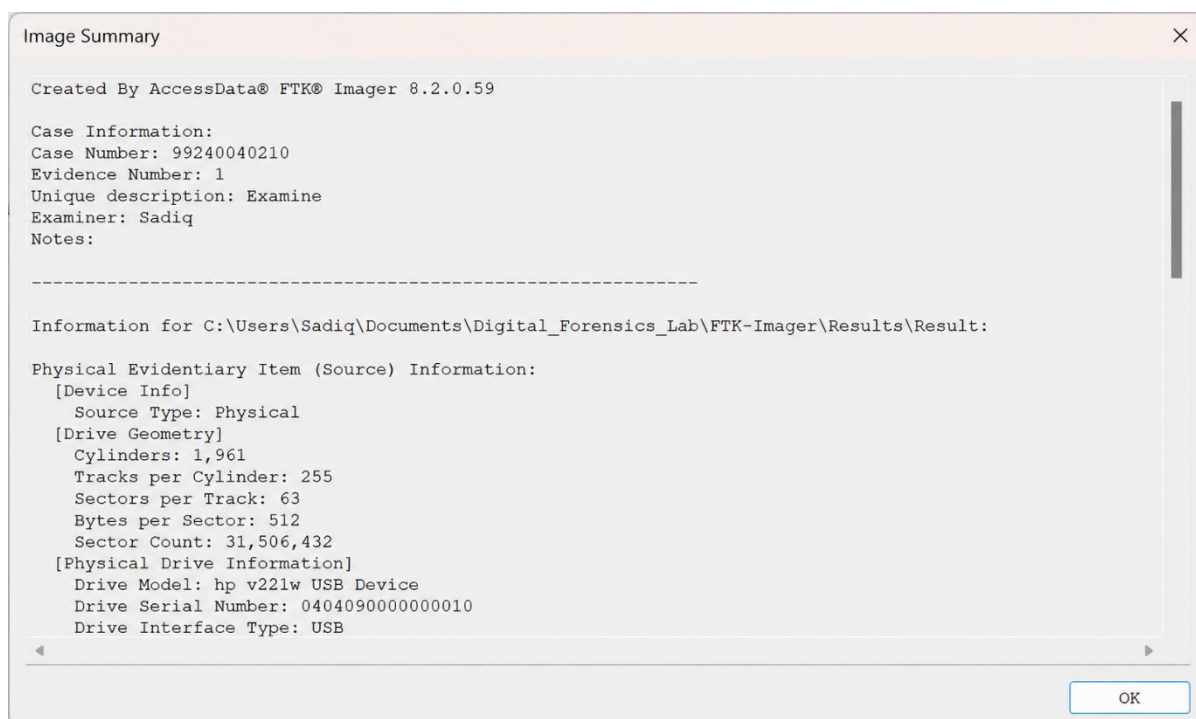


Figure 8: Image Summary detailing the physical device geometry

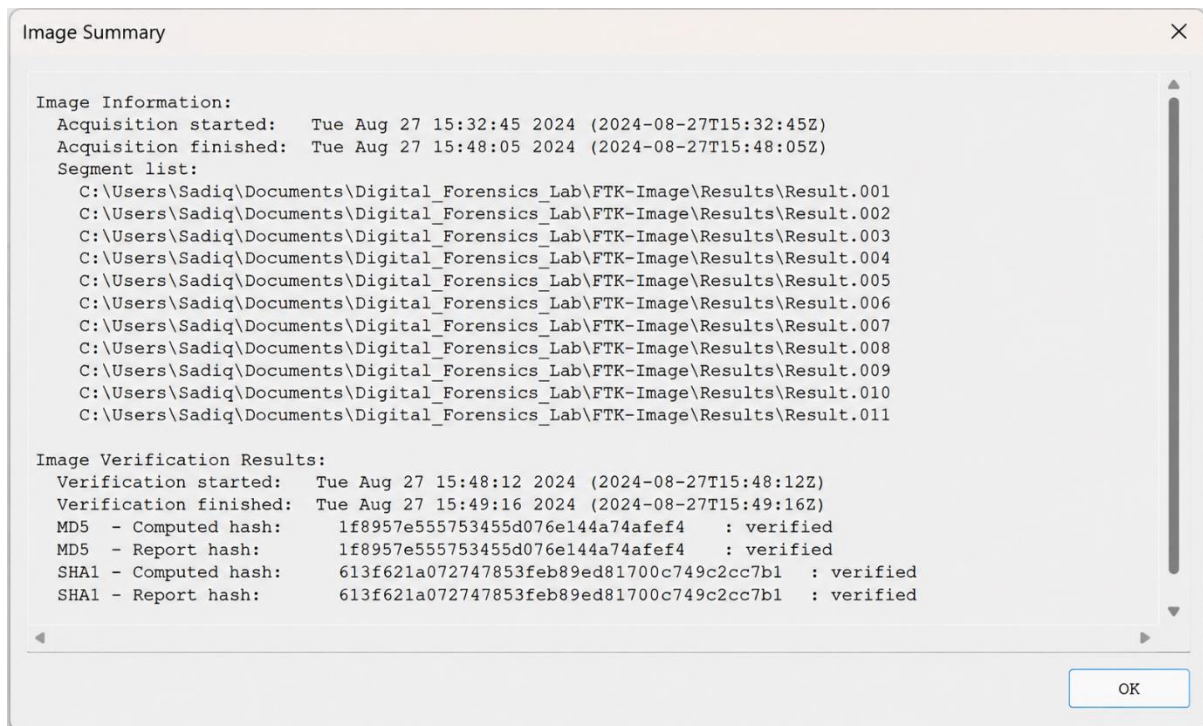


Figure 9: Image Summary detailing the hash verification results and segment list

Part B: Acquiring Volatile Memory (RAM) using WinPmem CMD

Step 1: Launch Command Prompt

Press Win + R, type cmd, and press Ctrl + Shift + Enter to open the Command Prompt as an **Administrator**.

Step 2: Navigate to Directory

Use the cd command to navigate to the folder containing the WinPmem executable:

DOS

A:

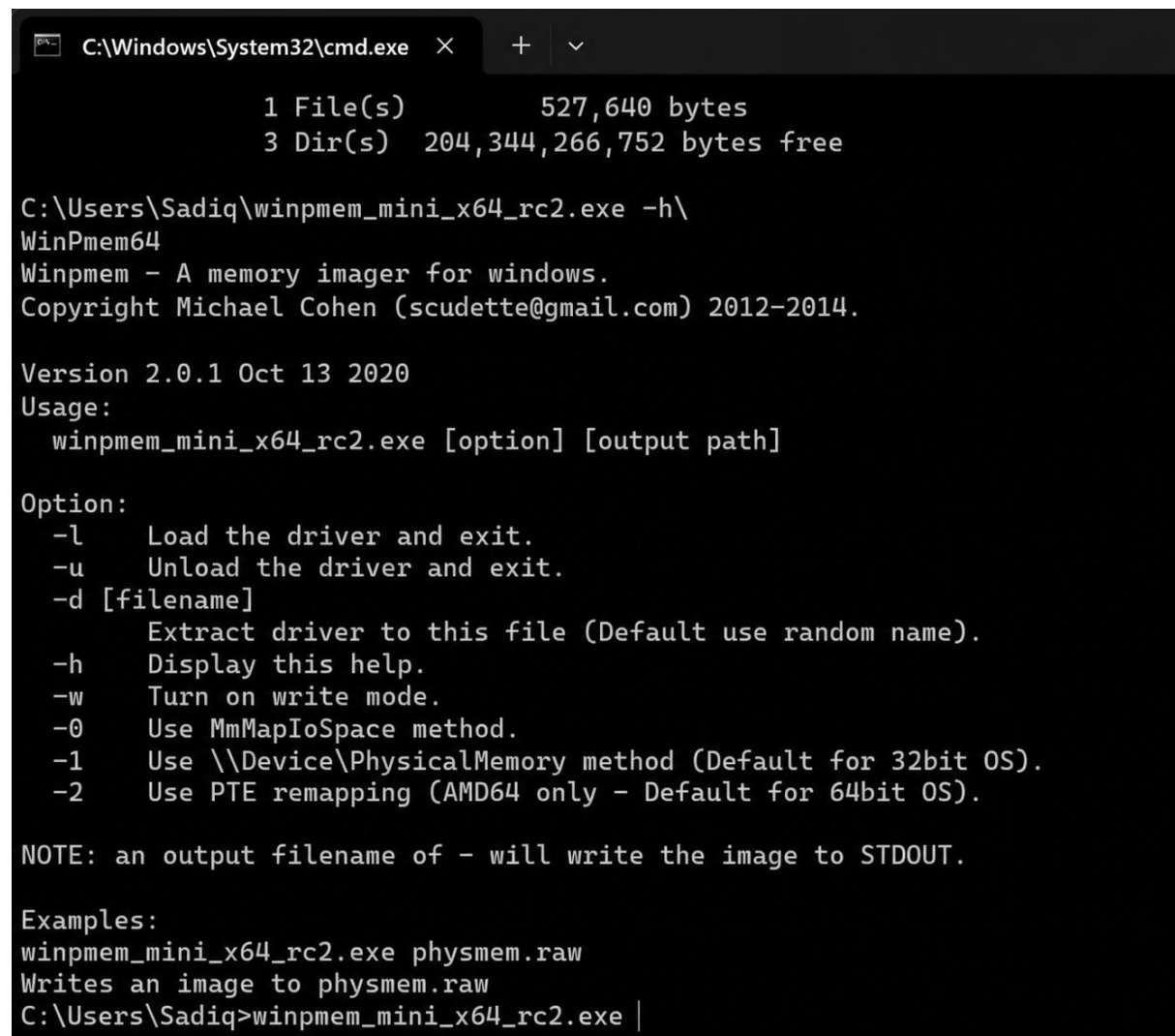
cd \DF

Step 3: Check Available Parameters

Execute the help command to view the WinPmem usage parameters:

DOS

winpmem_mini_x64_rc2.exe -h



```
C:\Windows\System32\cmd.exe  X + v

1 File(s)          527,640 bytes
3 Dir(s)  204,344,266,752 bytes free

C:\Users\Sadiq\winpmem_mini_x64_rc2.exe -h\
WinPmem64
Winpmem - A memory imager for windows.
Copyright Michael Cohen (scudette@gmail.com) 2012-2014.

Version 2.0.1 Oct 13 2020
Usage:
  winpmem_mini_x64_rc2.exe [option] [output path]

Option:
-l    Load the driver and exit.
-u    Unload the driver and exit.
-d [filename]
      Extract driver to this file (Default use random name).
-h    Display this help.
-w    Turn on write mode.
-o    Use MmMapIoSpace method.
-1    Use \\Device\PhysicalMemory method (Default for 32bit OS).
-2    Use PTE remapping (AMD64 only - Default for 64bit OS).

NOTE: an output filename of - will write the image to STDOUT.

Examples:
winpmem_mini_x64_rc2.exe physmem.raw
Writes an image to physmem.raw
C:\Users\Sadiq>winpmem_mini_x64_rc2.exe |
```

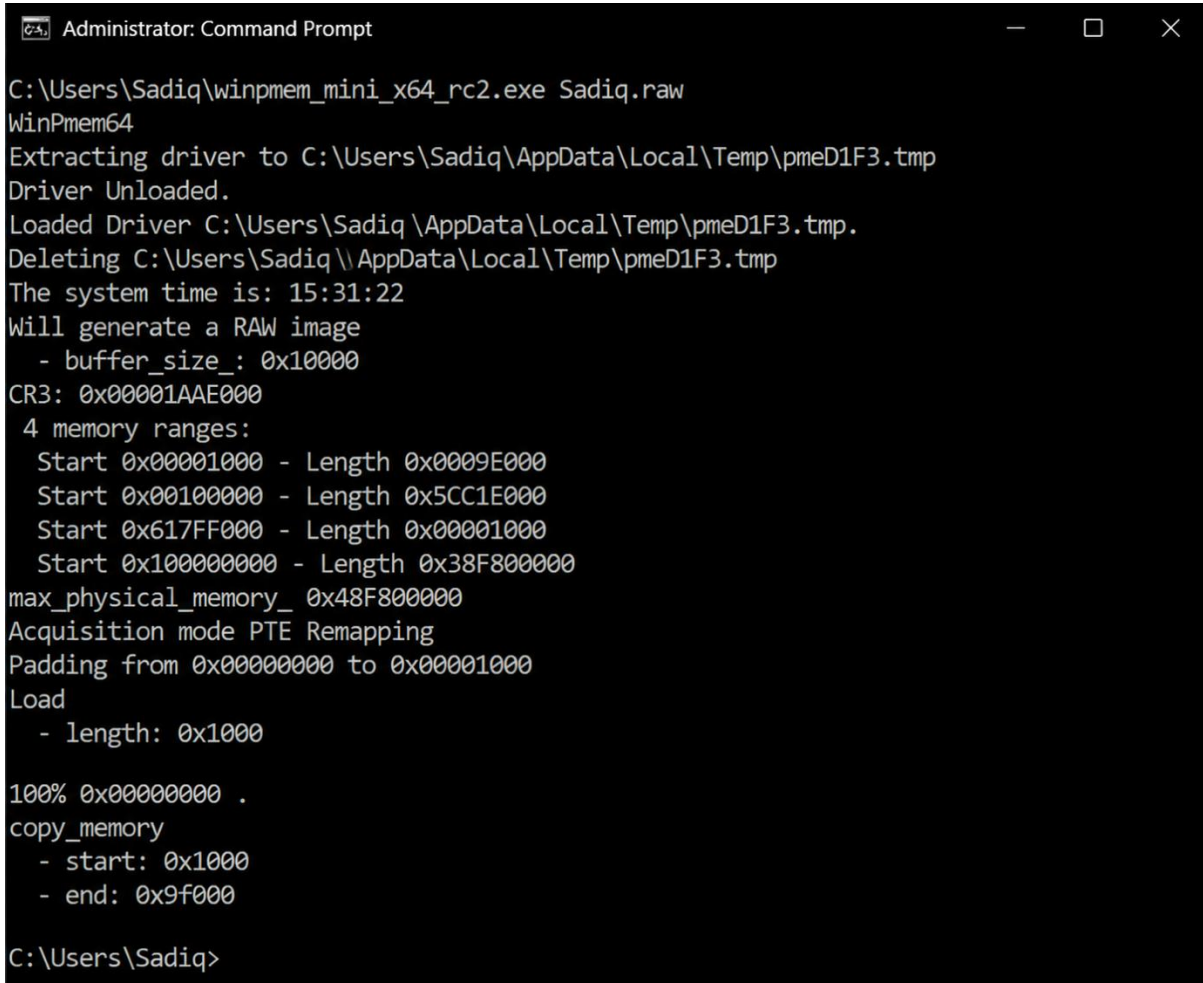
Figure 10: Viewing the WinPmem help menu and available parameters

Step 4: Execute Memory Capture Command

Run the memory extraction command, specifying the output filename `sadiq.raw`. The utility will map the physical memory and write it to the file:

DOS

```
winpmem_mini_x64_rc2.exe sadiq.raw
```



```
Administrator: Command Prompt

C:\Users\Sadiq>winpmem_mini_x64_rc2.exe Sadiq.raw
WinPmem64
Extracting driver to C:\Users\Sadiq\AppData\Local\Temp\pmeD1F3.tmp
Driver Unloaded.
Loaded Driver C:\Users\Sadiq\AppData\Local\Temp\pmeD1F3.tmp.
Deleting C:\Users\Sadiq\AppData\Local\Temp\pmeD1F3.tmp
The system time is: 15:31:22
Will generate a RAW image
  - buffer_size_: 0x10000
CR3: 0x00001AAE000
4 memory ranges:
  Start 0x00001000 - Length 0x00009E000
  Start 0x00100000 - Length 0x5CC1E000
  Start 0x617FF000 - Length 0x00001000
  Start 0x100000000 - Length 0x38F800000
max_physical_memory_ 0x48F800000
Acquisition mode PTE Remapping
Padding from 0x00000000 to 0x00001000
Load
  - length: 0x1000

100% 0x00000000 .
copy_memory
  - start: 0x1000
  - end: 0x9f000

C:\Users\Sadiq>
```

Figure 11: Initializing WinPmem and loading the extraction driver

```
Administrator: Command Prompt - winpmem_mini_x64_rc2.exe Sadiq.raw
Padding from 0x5CD1E000 to 0x617FF000
Load
- length: 0x4ae1000

97% 0x5CD1E000 .....
copy_memory
- start: 0x617ff000
- end: 0x61800000

98% 0x617FF000 .
Padding from 0x61800000 to 0x100000000
Load
- length: 0x9e800000

98% 0x61800000 .....
98% 0x61800000 .....
98% 0x61900000 .....
98% 0x61F80000 .....
copy_memory
- start: 0x100000000
- end: 0x48f800000

21% 0x100000000 .....XXXXX..X.....
26% 0x132000000 .....
30% 0x164000000 .....
34% 0x196000000 .....X.....
39% 0x1C8000000 .....X.....
43% 0x1FA000000 .....
47% 0x22C000000 .....
51% 0x25E000000 .....
```

Figure 12: Memory capture in progress

Step 5: Verification of Dump

Wait for the terminal to display Driver Unloaded. Navigate to the directory to verify the 18.2 GB sadiq.raw file.

```
copy_memory
- start: 0x100000000
- end: 0x48f800000

21% 0x100000000 .....XXXXX..X.....
26% 0x132000000 .....
30% 0x164000000 .....
34% 0x196000000 .....X.....
39% 0x1C8000000 .....X.....
43% 0x1FA000000 .....
47% 0x22C000000 .....
51% 0x25E000000 .....
56% 0x290000000 .....
60% 0x2C2000000 .....
64% 0x2F4000000 .....
69% 0x326000000 .....
73% 0x358000000 .....
77% 0x38A000000 .....
81% 0x3BC000000 .....
86% 0x3EE000000 .....
90% 0x420000000 .....
94% 0x452000000 .....
99% 0x484000000 .....
The system time is: 13:31:38
Driver Unloaded.
```

Figure 13: Successful completion of the volatile memory capture

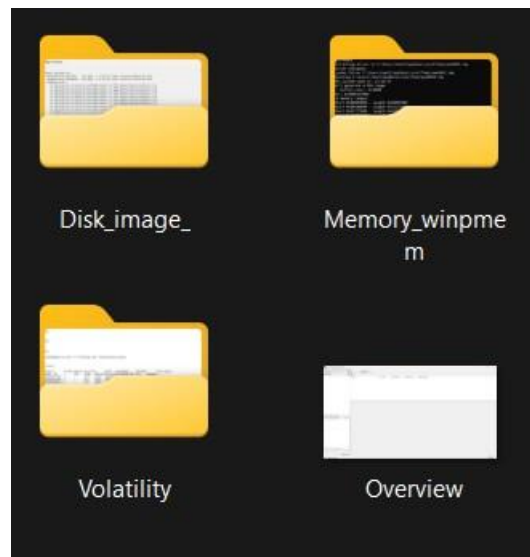


Figure 14: Organized forensic directories for Disk and Memory images

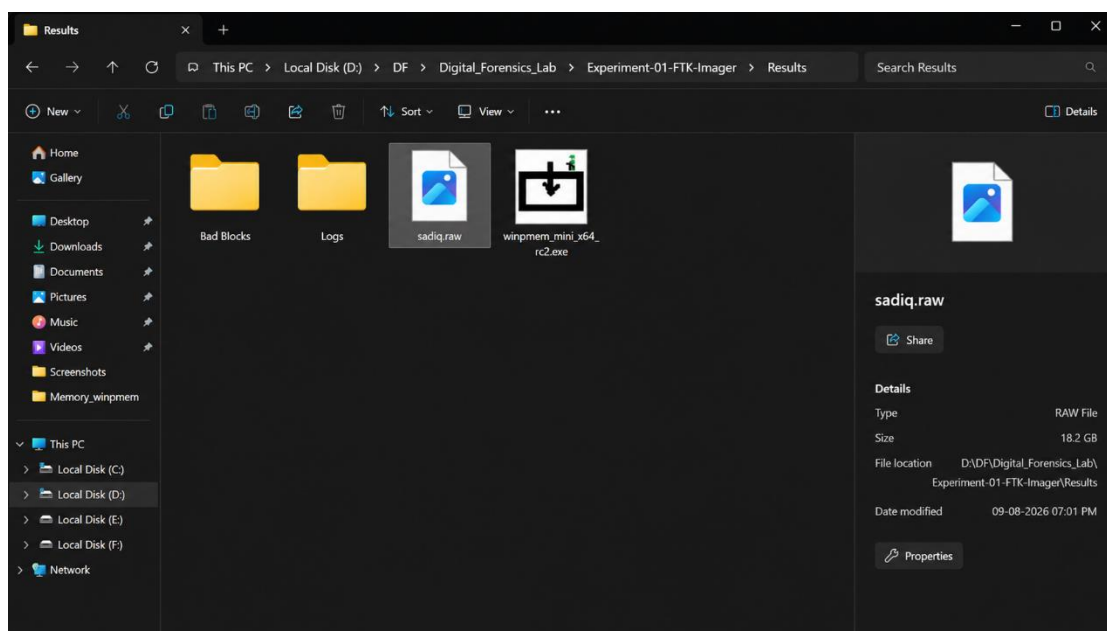


Figure 15: Verification of the generated sadiq.raw memory dump file

Part C: Volatile Memory Analysis using Volatility CLI / Workbench

Step 1: Launch Volatility Environment

Open PassMark Volatility Workbench (which acts as a GUI frontend for vol.exe).

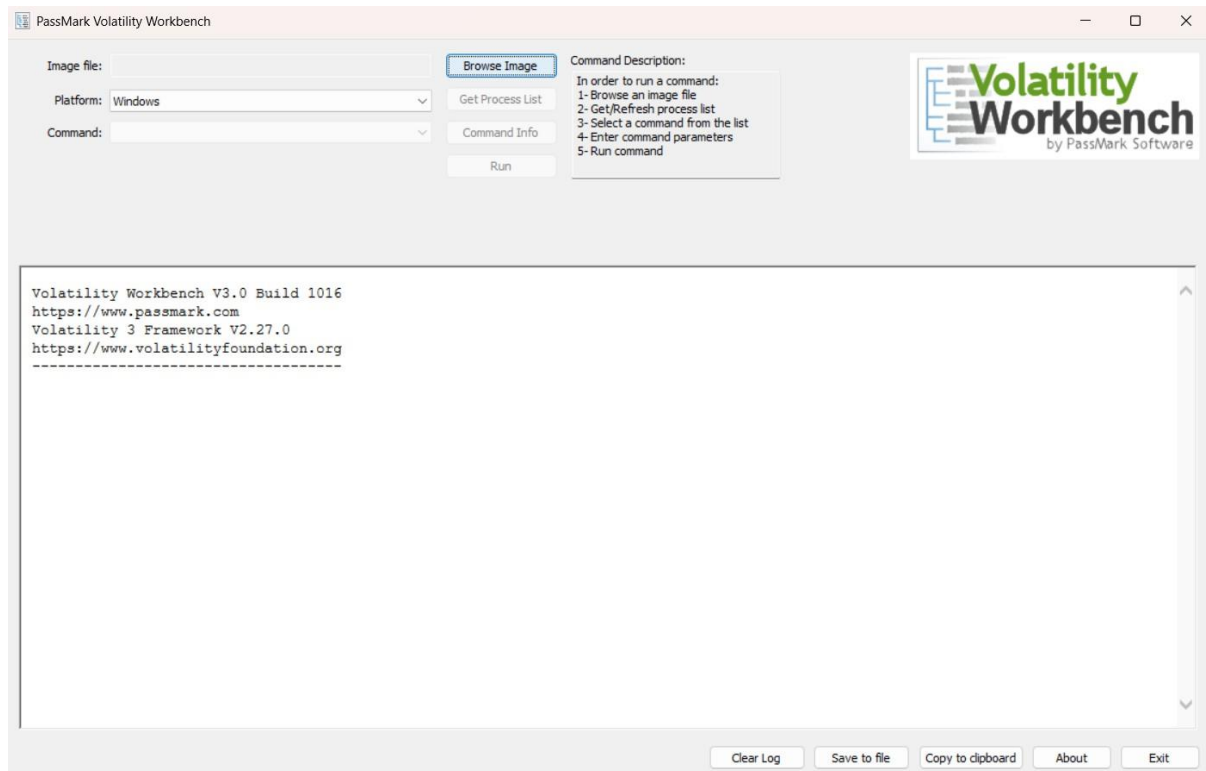


Figure 16: The default interface of PassMark Volatility Workbench

Step 2: Construct and Execute the Volatility Command

To parse the captured memory and list the active processes, load via **Browse Image**, select the **Windows** platform, choose windows.pslist.PsList from the dropdown, and click **Run**.

The console executes the following command in the background:

DOS

C:\Users\tamil\Downloads\VolatilityWorkbench\vol.exe" -f "A:\DF\sadiq.raw

Or

Use the workbench

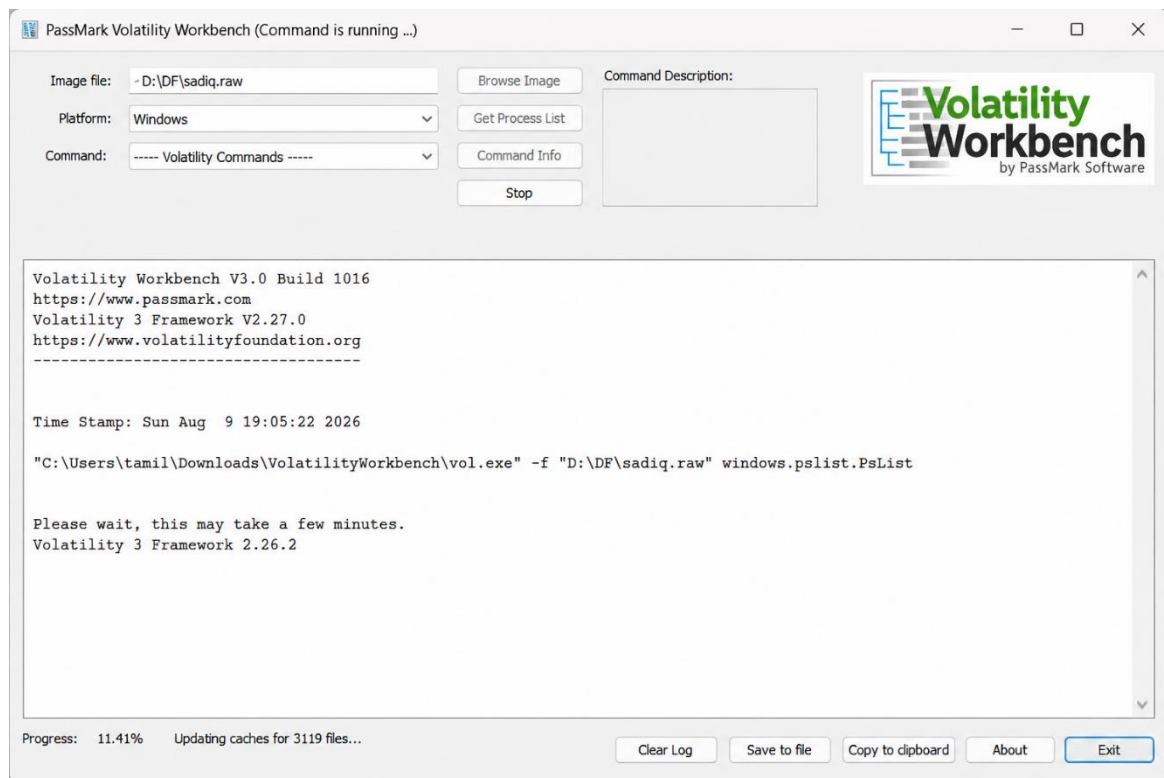


Figure 17: Execution of Volatility for the sadiq.raw image

5. OBSERVATIONS & RESULTS

1. FTK Imager Cryptographic Verification

FTK Imager generated a summary report mathematically confirming the non-volatile image data integrity:

- **MD5 Hash:** 1f8957e555753455d076e144a74afef4 : verified
- **SHA1 Hash:** 613f621a072747853feb89ed81700c749c2cc7b1 : verified

2. Memory Capture & CLI Parsing

The sadiq.raw file was successfully written using the winpmem_mini_x64_rc2.exe command. When analyzed, Volatility parsed the _EPROCESS blocks in kernel memory. The resulting

output log accurately detailed the Process IDs (PID), Parent Process IDs (PPID), Image File Names, and timestamps of all active programs:

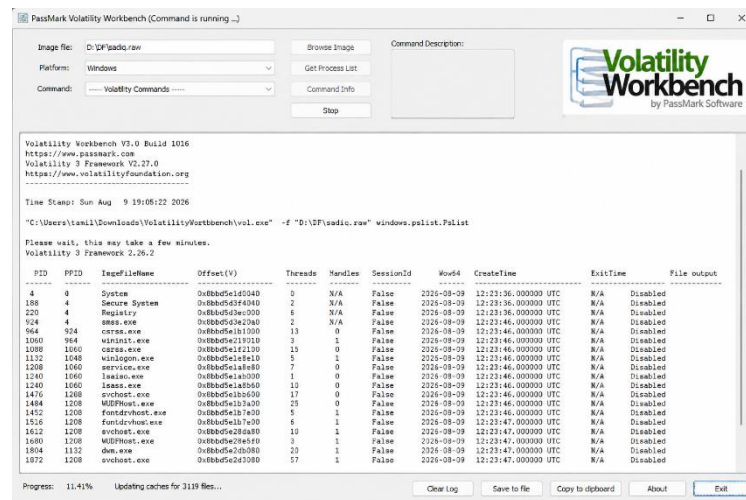


Figure 18: Volatility Workbench execution log displaying the extracted Process List

Note: The log accurately shows winpmem_mini_x (PID 19448) actively running, reflecting the acquisition tool executing at the time of the capture.

6. CONCLUSION

A complete digital forensics acquisition and analysis workflow was successfully executed via a combination of GUI and CLI tools. The target physical USB drive was securely imaged and verified using FTK Imager. Simultaneously, the system's live RAM was extracted via the Command Prompt using WinPmem. Finally, Volatility 3 command-line syntax was utilized to parse the raw memory file, successfully recovering critical live-state artifacts, including the process list and execution timestamps.